

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2022-23

Zoology**Paper: ZOB-502 — Functional Anatomy of Chordates**

Time: Three hours

Full Marks: 70

Note: Answer six questions including Question No. 1 which is compulsory. The figures in the right-hand margin indicate marks.

Question 1. Answer the following parts:

(a) Answer whether the given statements are True or False with 2-3 lines justification:
[5 × 2 = 10]

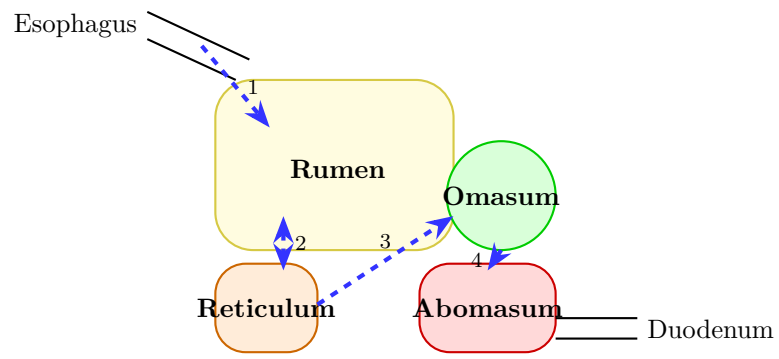
- (i) Cycloid scales are derived from lamellar bone component.
- (ii) Crescent shaped cusps are characterised by Lophodont dentition.
- (iii) Opisthonephros is also called head kidney due to its anterior position.
- (iv) The oviduct is modified Wolffian duct in vertebrates.
- (v) Mammalian lungs are structurally similar to avian lungs and have more efficient system of oxygen transport.

(b) Write in brief about the following: [5 × 2 = 10]

- (i) Labyrinthodont teeth
- (ii) Antlers
- (iii) Archinephros
- (iv) Hepatic portal system
- (v) Cerebral supremacy.

Question 2. Illustrate the general features of integument in aquatic vertebrates. How does the organisation of the integument in these organisms (such as sharks) allow lateral bending of the body without distorting the body shape? [10]

Question 3. With the help of a neat diagram, depict the different compartments of a ruminant stomach and their characteristics. Explain the mechanism behind the passage of food through these compartments. [10]



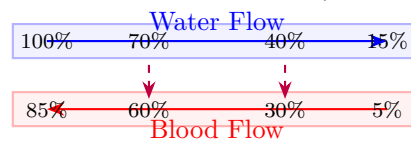
Question 4. Describe the evolution of jaws in gnathostomes highlighting the suitable modifications in structure and their advantages. [10]

Question 5. Write short notes on the following: [2 × 5 = 10]

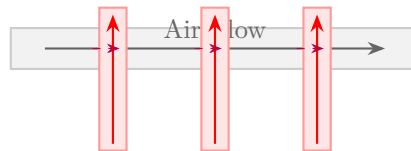
- (a) Lymph node
- (b) Neuromast organ

Question 6. Compare cross-current and counter-current mechanism of gaseous exchange with suitable examples and well-labelled diagram. [10]

A. Counter-Current Exchange (e.g., Fish Gills)

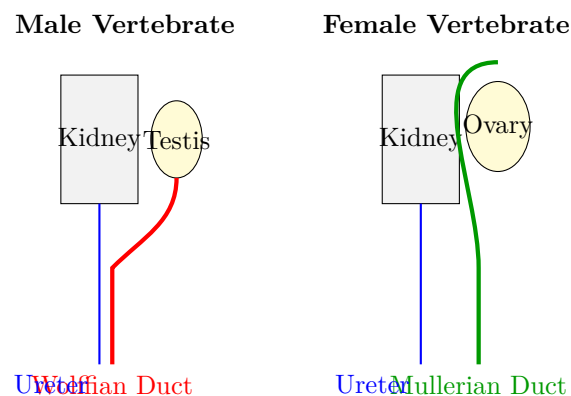


B. Cross-Current Exchange (e.g., Avian Lungs)



Question 7. What are aortic arches? How are they modified in Reptilia and Aves? Explain its evolutionary advantage in these classes. [10]

Question 8. Name and describe the functional kidney found in amniotes. Describe the urinogenital system and their ducts in male and female vertebrates with well-labelled diagram. [10]



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